

Standard Operating Procedure

Drop Ceiling Cat6 / Low-Voltage Cable Installation

Document No.	SOP-LV-001
Related Documents	SOP-LV-003, SOP-LV-006, SOP-LV-HSK-001
Effective Date	July 13, 2026
Revision	Rev. 0 (reconstructed)
Jurisdiction	Ontario (OHSA, O. Reg. 213/91) — Manitoba Workplace Safety and Health Act where applicable
Applies To	All field technicians installing Cat6 or other low-voltage cabling above suspended/drop ceilings
Approved By	_____

NOTE ON THIS DOCUMENT:

This SOP was reconstructed after the original file could not be located. It follows the same topic and structure as the original but is newly drafted — review carefully before relying on it for training or compliance purposes.

1. Purpose

This procedure establishes safe work practices for installing Cat6 and other low-voltage cabling above suspended (drop) ceilings, including ceiling void inspection, tile and grid access, cable routing and support, and coordination requirements at occupied commercial and institutional sites.

2. Scope

This SOP applies to all technicians performing cable installation, replacement, or service work that requires accessing the space above a suspended ceiling grid, at commercial, institutional, or school sites. It does not cover fall protection from ladders (see ladder safety procedure) or drilling into concrete floors or slabs (see SOP-LV-002).

3. Responsibilities

Site Supervisor

- Confirms the technician has reviewed available ceiling plans or as-builts for the work area before ceiling access begins.
- Coordinates ceiling access timing with the site/facility contact at occupied or school sites.
- Has authority to stop work if concealed services or hazards are identified above the ceiling.

Field Technician

- Inspects the ceiling void for existing services before running new cable.
- Follows cable routing and support requirements in Section 6.
- Reports any damaged tiles, grid, insulation, or suspected hazardous material immediately and stops work in that area.

4. Pre-Work Ceiling Void Inspection

Before running cable above a suspended ceiling, the technician must visually inspect the ceiling void at the work location for:

- Existing electrical conduit, BX cable, or junction boxes
- HVAC ductwork, diffusers, and associated controls
- Fire alarm and sprinkler piping or devices
- Plumbing lines or drains routed through the ceiling space
- Other existing low-voltage or communications cabling
- Insulation type and condition — treat unknown, loose-fill, or vermiculite-like insulation as a suspected hazardous material and stop work per Section 10

Where ceiling plans or as-builts are available, review them before removing tiles. Where the ceiling void has not been accessed before or plans are unavailable, treat the space as unverified and proceed with caution, checking tile-by-tile as work progresses.

5. Ceiling Tile & Grid Access

- Use a ladder positioned on a stable, level surface — never step directly on the ceiling grid or tiles, which are not rated to support a person's weight.
- Lift and remove tiles gently to avoid cracking; set removed tiles aside in a location where they will not be stepped on or damaged.
- Do not overload the grid with tools, cable reels, or materials — support weight from the ladder or a proper cable stand, not the grid itself.
- Replace all tiles in their original position and orientation before leaving the area, unless work is ongoing and the area is barricaded.

6. Cable Routing & Support

- Support cable using J-hooks or approved cable tray at intervals per manufacturer and code requirements (typically no greater than 4–5 ft) — do not lay cable directly on ceiling tiles or rest it on grid wires.
- Maintain minimum separation from electrical conduit and ballasts per the Canadian Electrical Code to avoid electromagnetic interference.
- Do not exceed the cable manufacturer's minimum bend radius when routing around obstructions.
- Label and dress cable runs neatly; avoid coiling excess slack loosely across the ceiling void.
- Where cable must cross HVAC ductwork or other services, route around — never rest cable on top of ductwork or diffusers.

7. Working Above Occupied Spaces & Schools

- Coordinate ceiling access timing with the school or facility contact, particularly in classrooms, hallways, and other high-traffic or occupied areas.
- Barricade or cone off the immediate work area below an open ceiling tile to prevent someone walking underneath while tiles are removed or tools are overhead.
- Never leave an open ceiling tile or ladder unattended in an area accessible to students, staff, or the public — replace the tile or secure the area during breaks.
- Keep noise and disruption to a minimum during instructional periods where required by the site agreement.

8. Fire-Rated Assemblies & Penetrations

Where a cable must penetrate a fire-rated ceiling, wall, or floor assembly, the penetration must be sealed with an approved firestop system rated for that assembly, per the National Building Code of Canada and local authority having jurisdiction requirements. Do not leave an unsealed penetration in a fire-rated assembly, even temporarily. Where the fire rating of an assembly is unknown, treat it as fire-rated until confirmed otherwise with the site contact.

9. Personal Protective Equipment

- Safety glasses or eye protection at all times when working above head height or handling ceiling tiles/insulation
- N95 (minimum) or P100 respirator when disturbing insulation or work areas with dust exposure
- Hard hat where overhead hazards exist or per site policy
- CSA-approved footwear and stable ladder setup per ladder safety procedure

10. Stop-Work Conditions

STOP WORK CONDITION:

Do not continue work in a ceiling void where insulation is unknown, loose-fill, or vermiculite-like in appearance. Treat as suspected asbestos-containing material, stop work at that location, and escalate to the Site Supervisor before proceeding.

- Any suspected hazardous material (insulation, pipe wrap, ceiling tile) is encountered
- An unidentified or unmarked service is found and cannot be confirmed safe to work around
- Ceiling grid, tiles, or structure show signs of damage or instability

11. References

- Ontario Occupational Health and Safety Act (OHSA) and O. Reg. 213/91 — Construction Projects
- O. Reg. 278/05 — Designated Substances – Asbestos on Construction Projects and in Buildings and Repair Operations
- Canadian Electrical Code (CEC), Part I
- National Building Code (NBC) of Canada, applicable sections on firestopping
- Manitoba Workplace Safety and Health Act and Regulation, where applicable
- SOP-LV-003 — Ontario Jobsite Health & Safety Checklist
- SOP-LV-006 — Utility Strike Prevention
- SOP-LV-HSK-001 — Jobsite Housekeeping

12. Technician Sign-Off

By signing below, the technician confirms they have read, understood, and agree to follow this procedure when installing low-voltage cabling above suspended ceilings.

Printed Name	Signature	Date